

Online Appendix

Quality Regulation and Competition: Evidence from Pharmaceutical Markets

Juan Pablo Atal, José Ignacio Cuesta and Morten Sæthre

List of Figures

A.1 Labeling of bioequivalent drugs	19
A.2 Innovator and branded generics price premiums relative to unbranded generics . .	20
A.3 Policy variation induced by bioequivalence requirements	21
A.4 Survey results for perceived quality	22
A.5 Correlation between drug exit after regulation and baseline revenue	23
A.6 Welfare effects under alternative assumptions for non-bioequivalent drugs' valuation	24
A.7 Heterogeneity in the effects of quality regulation and the role of certification costs . .	25
A.8 The impact of a quality standard and quality disclosure on variable profits	26
A.9 Number of recalls per month	27
A.10 Consumer survey elicitation of perceived quality and price	28
A.11 Drug expenditures and income	29

List of Tables

A.1 Timing of reform to quality regulation	30
A.2 Descriptive statistics	31
A.3 First stage of demand estimation	32
A.4 Effects of quality regulation on proxies of drug quality	33
A.5 Summary statistics from consumer survey data	34
A.6 Decomposition of the effects of quality regulation on drug prices	35
A.7 Effects of quality regulation on number of drugs by drug attributes	36
A.8 Effects of quality regulation on drug prices by drug attributes	37

A Decomposition of Price Effects

The effects of quality regulation on average prices combine three channels: Drug-specific price changes, changes in shares, and changes in the composition of drugs in each market. To understand the drivers of price effects, we develop a decomposition of the evolution of the price index into such components.¹

We define $\hat{P}_{mt} \equiv \sum_{i \in \mathcal{I}_{mt}} w_{it} P_{it}$ as the share-weighted average of log prices in market m and month t . Denote the set of drugs in the market in t that were also in the market in the baseline period as $\mathcal{S}_{m,t} \equiv \mathcal{I}_{mt} \cap \mathcal{I}_{m0}$; the set of drugs that entered market m after the baseline period and remain in the market in t as $\mathcal{E}_{mt} \equiv \mathcal{I}_{mt} \setminus \mathcal{I}_{m0}$; and the set of drugs that exited between the baseline period and t as $\mathcal{X}_{mt} \equiv \mathcal{I}_{m0} \setminus \mathcal{I}_{mt}$. We then decompose the change in the share-weighted average of log prices between a baseline period $t = 0$ and any period $t > 0$ as

$$\begin{aligned} \hat{P}_{mt} - \hat{P}_{m0} = & \underbrace{\sum_{i \in \mathcal{S}_{mt}} w_{i0}(P_{it} - P_{i0})}_{\Delta P_{mt,PC}} + \underbrace{\sum_{i \in \mathcal{S}_{mt}} (P_{i0} - \hat{P}_{m0})(w_{it} - w_{i0})}_{\Delta P_{mt,RW}} \\ & + \underbrace{\sum_{i \in \mathcal{S}_{mt}} (w_{it} - w_{i0})(P_{it} - P_{i0})}_{\Delta P_{mt,CS}} + \underbrace{\sum_{i \in \mathcal{E}_{mt}} w_{it}(P_{it} - \hat{P}_{m0})}_{\Delta P_{mt,E}} - \underbrace{\sum_{i \in \mathcal{X}_{mt}} w_{i0}(P_{i0} - \hat{P}_{m0})}_{\Delta P_{mt,X}}. \end{aligned}$$

The first term, $\Delta P_{mt,PC}$, is the change in the share-weighted average price due to price changes among incumbent drugs, while holding the weights fixed at their baseline level. The second term, $\Delta P_{mt,RW}$, is the change in the share-weighted average due to changes in market shares, while holding prices fixed. This term is positive when relatively expensive incumbent drugs increase their market share. The third term, $\Delta P_{mt,CS}$, is the change in share-weighted prices due to the correlation between price changes and changes in market shares. This term is positive when drugs that increase their prices also increase their market shares. The fourth term, $\Delta P_{mt,E}$, captures price changes due to the entry of drugs into the market. This component is positive whenever drugs that enter the market are more expensive than the average drug in the baseline period. Finally, the fifth term, $\Delta P_{mt,X}$, measures the price change due to the exit of drugs. This component is positive whenever drugs that exit the market are less expensive than the average drug in the baseline

¹Nevo and Hatzitaskos (2006) use the same price index (a weighted average of log prices) and study the extent to which changes in the composition (changes in weights) can explain changes in average prices between high- and low-demand periods. However, they do not provide an explicit decomposition, nor do they explicitly include the role of entry and exit. Moreover, we are not aware of other papers that employ the same strategy to decompose the effects of an event of interest (e.g., a reform or changes in demand) into its sources. This decomposition is analogous to decompositions of productivity changes over time into within-plant changes and changes in the composition of plants over time, including entry and exit (Foster *et al.*, 2001).

period. Therefore, the price index can be decomposed as

$$\hat{P}_{mt} = \hat{P}_{m0} + \Delta P_{mt,PC} + \Delta P_{mt,RW} + \Delta P_{mt,CS} + \Delta P_{mt,E} + \Delta P_{mt,X}. \quad (\text{A.1})$$

We estimate the effect of quality regulation on each component by estimating equation (2) using $\hat{P}_{mt,PC} \equiv \hat{P}_{m0} + \Delta P_{mt,PC}$, $\hat{P}_{mt,RW} \equiv \hat{P}_{m0} + \Delta P_{mt,RW}$, $\hat{P}_{mt,CS} \equiv \hat{P}_{m0} + \Delta P_{mt,CS}$, $\hat{P}_{mt,E} \equiv \hat{P}_{m0} + \Delta P_{mt,E}$ and $\hat{P}_{mt,X} \equiv \hat{P}_{m0} + \Delta P_{mt,X}$ as dependent variables. The sum of the OLS coefficients on T_{mt} from these regressions equals the coefficient on T_{mt} when estimating equation (2) for $\hat{P}_{mt}$.

Table A.6 displays estimates of the effects on each price index component, across and within drug segments. Overall, the results reveal that more than half of the price increases come from price changes among incumbents, which supports the interpretation that drug exit reduced the intensity of price competition. For the same reason, when quality is revealed, new entrants are expected to have prices that are higher than pre-reform average prices. Our decomposition confirms this, as around a third of the increase is due to compositional changes, in particular to the entry of more expensive drugs.

B Effects on Drug Quality

In this section, we provide supplemental evidence on whether the bioequivalence regulation affected the observable proxies of quality for drugs in the market. The effects of the reform on quality are key to understanding its welfare consequences. Unfortunately, direct measures of quality (e.g., results from drug testing) are not available in our setting, so we use adverse health events associated with specific drugs and drug recalls as indirect measures. Note that both outcome variables are mostly related to drug safety—which was already covered by the quality standards in place before the reform we study—whereas bioequivalence certification primarily targets drug efficacy. We view this analysis as a complement to our primary empirical strategy in Section 6, where we estimate the changes in quality resulting from the regulation using a structural model.

Let the quality outcome for market m at time t be $y_{mt} = \mu_{mt} \text{sales}_{mt}$, where μ_{mt} is the probability of an adverse effect associated with drugs in market m , and sales_{mt} are sales of drugs in m . Similarly to Jin and Leslie (2003), we model the probability of an adverse outcome—either an adverse health event or a drug recall—as $\mu_{mt} = \mu_{m0} + \gamma_t + \theta T_{mt} + \varepsilon_{mt}$, which combines a baseline probability μ_{m0} , with time shocks common to all markets γ_t , a shifter related to quality regulation θT_{mt} , and a random shock ε_{mt} . This simple framework motivates the estimating equation

$$\frac{y_{mt}}{\text{sales}_{mt}} = \mu_{m0} + \gamma_t + \theta T_{mt} + \varepsilon_{mt}, \quad (\text{A.2})$$

where θ measures the effect of stronger quality regulation on the number of adverse outcomes per unit of sales, whereas market and time fixed effects capture μ_{m0} and γ_t .

B.1 Evidence from Adverse Health Events

A first set of outcomes related to quality are the adverse health events associated with drug consumption. We collected data on yearly clinical outcomes between 2010 and 2017 for ICD-10 diagnosis codes related to molecules in our sample. The data consist of public records compiled by DEIS (2019), which encompass admissions, hospitalization days, and the number of surgeries across all hospitals in Chile. We link diagnoses to molecules using a crosswalk between the American Hospital Formulary Service (AHFS) and ICD codes, which tracks adverse health events associated with the consumption of drugs (WHO, 2007).²³ Our analysis focuses on the 71 molecules with at least one listed adverse effect.⁴ In our setting, these events are rare. In 2010, there were on average 7.3 admissions, 13.2 hospital days, and 0.002 surgeries per 100,000 daily doses sold across all markets.

We estimate equation (A.2) for these outcomes. Columns 1–3 in Table A.4-A display the results across all markets. We find no evidence suggesting that stronger quality regulation decreased the number of discharges and the number of days associated with them. Moreover, we find no evidence of heterogeneous effects on these outcomes across small and large markets in Table A.4-B. These results suggest that stronger quality regulation was not able to reduce the adverse health effects of drugs.

B.2 Evidence from Drug Recalls

To study the effects of quality regulation on drug recalls, we collected data on the 209 recalls of prescription drugs that occurred during our study period. ISP implements recalls as preventative sanitary measures upon notice of adverse events linked to licensed drugs.⁵ In the period we cover, there is an average of 2.2 recalls per month, of which 1.4 relate to molecules without bioequivalence requirements, and 0.8 relate to molecules with bioequivalence requirements.⁶

Our estimates of equation (A.2) in the sample of molecules with bioequivalence requirements

²³When several ICD codes capture adverse effects related to the same molecule, we assign outcomes to molecules using weights for sales volume across molecules within each ICD code.

⁴As an example, admissions coded under “T455 - Poisoning by, adverse effect of and underdosing of anticoagulants and antithrombotic drugs” are attributed to the consumption of Acenocoumarol, an anticoagulant.

⁵The results from the regressions on market outcomes are very similar when restricted to this sample. The results are available from the authors upon request.

⁶The reasons for these recalls can be categorized broadly into manufacturing defects, including chemical defects and contamination (71%); efficacy concerns or side effects (9%); or others, which mostly correspond to counterfeit drugs or mislabeling (20%). Due to the small number of recall events, we use all data irrespective of the specific reason.

⁶Figure A.9 shows the monthly frequency of recalls, split into drugs with bioequivalence requirements in our sample and drugs without a bioequivalence requirement. We cannot reject the hypothesis of a common trend in recalls over time across these two groups. We estimate an OLS regression for recall rates on an indicator for requirements and its interaction with a time trend and find that the coefficient on the latter is not statistically different from zero.

provide no evidence suggesting that stronger quality regulation improved drug quality as measured by recalls. Column 4 in Table A.4-A displays the results across all markets, while Table A.4-B does so by market size. Our point estimates are close to zero across both specifications. These results suggest that imposing bioequivalence requirements did not help decrease drug recalls relative to the quality standards previously in place.

C Additional Heterogeneity Analysis by Drug Characteristics

In this section, we expand on the analysis of the reform's effects by investigating further dimensions of heterogeneity. Specifically, we are interested in evaluating the impact of the reform on different types of drugs. We split markets along two dimensions of interest: whether they treat a chronic condition, following the classification in Huber *et al.* (2013), and whether they are classified as having a high impact on mortality, following the classification in Chandra *et al.* (2024).

In our sample, there are 115 markets, of which 50 treat a chronic condition and 19 are classified as having a high impact on mortality. We estimate two versions of equation (2), allowing for heterogeneous effects of the reform along these dimensions. The results are shown in Tables A.7 and A.8 for the effects of the policy change on the number of drugs and prices, respectively.

Considering heterogeneity by whether drugs treat a chronic condition has a direct link to our modeling framework: a key aspect of quality regulation is to provide information, and consumers in markets for drugs targeting chronic conditions are likely more informed about the quality of generics at baseline. Therefore, we should expect stronger effects of the regulation in markets for non-chronic diseases. Reassuringly, our results align with this prediction. Overall, we find that the effects on the number of drugs are more pronounced for generics used in the treatment of non-chronic diseases. Moreover, we find statistically significant differences in price effects between the two types of drugs among unbranded generics.

In contrast, considering heterogeneity by whether a drug has high mortality impacts is not directly linked to our model. However, estimating heterogeneous impacts along this dimension is relevant in its own right. Overall, we find that quality regulation had similar effects on the number of drugs for markets with high and low mortality effects. Conversely, we find that price increases are more pronounced for markets for drugs with higher mortality effects. These results suggest that the adverse effects of quality regulation in terms of price increases for generics are concentrated among markets for drugs with high mortality impacts.

D Description of Consumer Survey

D.1 Methodology and Results

To aid the interpretation of our descriptive evidence and guide our model assumptions, we collected additional survey data through interviews with consumers to gather information on

perceived quality, perceived price differences, the relationship between physician prescription behavior and consumer choices, and other characterization variables. Section D.2 below displays the questionnaire.

A surveying team composed of six members conducted surveys in four counties in the city of Santiago, namely Ñuñoa, Providencia, Puente Alto, and Santiago. Surveyors recruited consumers outside pharmacies within such counties, where consumers were purchasing drugs. This recruiting strategy aimed at constructing a sample of consumers familiar with the pharmaceutical market. Recruited participants were asked to participate in a survey with a duration of between 5 and 10 minutes, and were offered no compensation.

To collect data on perceived quality and price differences, we focus on a particular market, Atorvastatin, a molecule commonly prescribed as a treatment for high cholesterol. Within that market, we focus on four relevant drugs. In particular, we work with (i) a popular innovator drug called Lipitor, which is produced by Pfizer, (ii) a bioequivalent branded generic called Lipoten, produced by Pharmavita, (iii) a bioequivalent unbranded generic called simply Atorvastatina, produced by Mintlab, and (iv) a non-bioequivalent unbranded generic also called Atorvastatina and produced by Mintlab. For reference, the prices of these drugs in the market are around \$50,000 CLP, \$10,000 CLP, \$2,500 CLP, and \$2,500 CLP, respectively (\$71.4, \$14.3, \$3.6, and \$3.6 USD, respectively). We elicit perceived quality using a paper sheet that showed the four drugs, which Figure A.10 displays. For these questions, we use a 1-7 scale—this is a natural scale for Chileans as it is used as the standard scale for school exam scores.

The final sample includes $N = 401$ consumers. Table A.5 provides summary statistics for the main variables in the survey. Among consumers in the sample, 58% report having a household member with a chronic disease, and 34% report purchasing Atorvastatin for a household member. In terms of purchase behavior, 41% often purchase innovator drugs, 20% often purchase branded generics, and the remaining 39% often purchase unbranded generics. The results of the survey and their relationship to our main analysis are discussed in Section 4. We code observations in which a consumer answered “I don’t know” or “I don’t recall” as missing.

D.2 Questionnaire

We are conducting a survey about the quality perception of drugs sold in pharmacies. We will ask you a few questions regarding the quality and prices of drugs. In all examples, we will focus on a drug called Atorvastatin, which is commonly used to control cholesterol levels. While we understand that it may be that no one in your household takes Atorvastatin, we ask that you consider it as an example and think as if you had to acquire it for a family member.

(Show pictures of four drugs) Consider a scale of 1 to 7, where 1 is a drug of the minimum quality and that does not have the desired therapeutic effects and 7 is a drug of the highest

quality that has exactly the expected therapeutic effects. What level of quality do you think the following drug has?

- Innovator
- Bioequivalent unbranded generic
- Unbranded generic
- Bioequivalent branded generic

E Empirical Model Details

E.1 The Constant Expenditure Model

Consider the direct utility function for consumer i over quantities q_{ij} of products indexed by j with quality index ψ_{ij} and the amount z_i of the numeraire

$$u(\sum_j \psi_{ij} q_{ij}, z_i) = \varphi \ln(\sum_j \psi_{ij} q_{ij}) + (1 - \varphi) \ln z_i,$$

where the quality index is specified as

$$\psi_{ij} = \exp\left(\frac{\delta_j + \epsilon_{ij}}{\alpha}\right),$$

with δ_j being a utility term common across consumers for product j , while ϵ_{ij} is an individual-specific random utility term. This utility function implies that the goods indexed by j are perfect substitutes, and the consumer will choose the one with the highest ratio of quality over price ψ_{ij}/p_j . The consumer has Cobb-Douglas preferences over z_i and $\sum_j \psi_{ij} q_{ij}$, implying constant budget shares $1 - \varphi$ and φ allocated to z_i and the products indexed by j . The form of the quality index implies that α governs the weight on log price relative to the quality components in product choice, which is reflected in the fact that maximizing ψ_{ij}/p_j is equivalent to maximizing $\delta_j - \alpha \ln p_j + \epsilon_{ij}$.

The conditional direct utility function when choosing product j (i.e., when $q_{ij} > 0$ and $q_{i,k \neq j} = 0$) can be written as

$$u_{ij} = u(\psi_{ij} q_{ij}, z_i) = (1 - \varphi) \ln z_i + \varphi \ln q_{ij} + \varphi \ln \psi_{ij},$$

which implies that quantity q_{ij} and the quality components— δ_j and ϵ_{ij} from $\ln \psi_{ij}$ —are additively separable in utility for a given choice j . Maximizing the conditional direct utility subject to a budget constraint $p_j q_{ij} + z_i \leq y_i$ gives the conditional ordinary demand functions

$$\begin{aligned} q_{ij}(p_j, y_i) &= \frac{\varphi y_i}{p_j} \\ z_i(p_j, y_i) &= (1 - \varphi) y_i. \end{aligned}$$

Inserting into the conditional direct utility, we obtain the conditional indirect utility function

$$\tilde{v}_{ij} = \ln y_i - \varphi \ln p_j + \frac{\varphi}{\alpha}(\delta_j + \epsilon_{ij}),$$

which, after rescaling by α/φ , becomes

$$v_{ij} = \frac{\alpha}{\varphi} \ln y_i - \alpha \ln p_j + \delta_j + \epsilon_{ij},$$

where the consumer chooses the product $j^* = \arg \max_j v_{ij}$, while quantities demanded conditional on this choice are given by the conditional ordinary demand functions above and zero for all non-chosen products.⁷ Specifying δ_j as a function of observed and unobserved product characteristics, and appropriately redefining the random utility term, we obtain the utility function in Section 6.1.

Relationship between income and expenditures. As noted in Section 6.1, the functional form in income and price of this demand model implies that consumers allocate a constant income share φ_{mt} to the market, which is spent on a single preferred alternative. We test this model prediction using auxiliary data from the 2016 National Household Spending Survey (INE, 2018). For each household h , we construct income per capita y_h as well as drug expenditure e_h . Figure A.11 shows a binned scatter plot relating the two variables as well as a histogram of y_h . We find that the relationship between income and expenditures is almost linear along most of the distribution of per-capita income.

E.2 Market Size Approximation

In our demand model, we normalize the utility of the outside good to zero and model the expenditure share of each product as a function of observable characteristics. From observed data on quantities and prices, one needs to take a stance on the market size in terms of total potential budget spent in market m at time t , $B_{mt} \equiv \varphi_{mt} Y_t$. This variable plays an analogous role in our discrete-continuous demand model to quantity-based market size in the commonly used unit-demand models.

Since the potential budget is not observed and we do not have any measures that would allow us to infer it directly, such as the number of individuals with a diagnosis for which a given drug would be relevant, we approximate B_{mt} for each ATC-5 code m and year t using the insights of Huang and Rojas (2013, 2014). Specifically, we adapt the procedure used by Dubois and Lasio (2018) to the specific model we use. We start by specifying a simple logit version of our model,

⁷Note that we ignore the constant $\varphi \ln \varphi + (1 - \varphi) \ln(1 - \varphi)$, which is common across all alternatives in the conditional indirect utility function.

where we use the expenditure share equations

$$\ln s_{jmt} - \ln s_{0mt} = v_{jmt}^k - \alpha \ln p_{jmt} + x'_{jmt} \beta + \xi_{jmt}, \quad s_{jmt} = \frac{p_{jmt} q_{jmt}}{B_{mt}},$$

for two observed, distinct goods j and j' to derive

$$\ln p_{jmt} q_{jmt} - \ln p_{j'mt} q_{j'mt} = v_{jmt}^k - v_{j'mt}^{k'} - \alpha (\ln p_{jmt} - \ln p_{j'mt}) + (x_{jmt} - x_{j'mt})' \beta + \xi_{jmt} - \xi_{j'mt}, \quad (\text{A.3})$$

which does not depend on the potential budget. The parameters of this equation can be consistently estimated by 2SLS using similar instruments to those introduced in Section 6.1. In a second step, we use the equation

$$\ln p_{jmt} q_{jmt} - \ln p_{0mt} q_{0mt} = v_{jmt}^k - \hat{\alpha} \ln p_{jmt} + x'_{jmt} \hat{\beta} + \xi_{jmt}, \quad (\text{A.4})$$

where we take the parameters estimated from equation (A.3) as given, and $\ln p_{0mt} q_{0mt}$ is the log of the expenditure allocated to the outside good, which is common for all products j in market m in year t . Note that the potential budget by definition is equal to the expenditure on all observed alternatives in addition to the expenditure on the outside good, therefore knowing the latter is equivalent to knowing the potential budget with typical price-quantity data.⁸ We specify expenditures on the outside good as

$$\ln p_{0mt} q_{0mt} = \sum_{r=0}^4 \varsigma_{rmt} t^r + \varsigma_p \overline{\ln p_{mt}} + \bar{x}'_{mt} \varsigma_x, \quad (\text{A.5})$$

where the first sum is a fourth-degree polynomial in time (including a constant) that is allowed to differ across markets, and $\overline{\ln p_{mt}}$ and $\bar{x}_{mt}$ are the average log price and average characteristics in market m and year t . The average terms capture the effect of the attractiveness of the inside goods in competition with the outside good. This equation is estimated using a modified GMM-IV procedure, minimizing

$$O(\varsigma) = G(\varsigma)' W G(\varsigma) + \varsigma' \Lambda \varsigma,$$

where $G(\varsigma) = Z' \xi(\varsigma)$ is the vector of moments, Z is a matrix of the exogenous variables in equation (A.5) including price in Norway for the corresponding ATC and year, and W is the weighting matrix for the moments. The quadratic form in the parameter vector ς is a regularization term, with Λ as a diagonal matrix of regularization parameters. We allow two regularization parameters: one for parameters controlling the market-time polynomials, and one for parameters controlling the effect of average characteristics. We add regularization due to the large number of parameters in equation

⁸It can be noted that only the aggregate expenditure on the outside good is important in this model, meaning that no assumptions on p_{0mt} are necessary.

(A.5), and calibrate the regularization parameters to minimize RMSE using cross-validation.

E.3 Aggregation of Unbranded Drugs for Demand Estimation

Since the IQVIA sales data only report the sales of unbranded generics at the aggregate segment level, we do not observe sales for individual unbranded products. Let the number of unbranded generic products that are active in market m at time t be denoted by N_{mt}^u , and the set of such products by $\mathcal{J}_{mt}^u$. More generally, let the set of active products in segment k in market m at time t be denoted by $\mathcal{J}_{mt}^k$.

Given this data limitation, we use the average expenditure share of unbranded generics

$$s_{mt}^u \equiv \frac{1}{N_{mt}^u} \sum_{j \in \mathcal{J}_{mt}^u} s_{jmt},$$

as the observed expenditure share for the segment. This approach implicitly uses information only on the choice of the unbranded generic segment as a whole, rather than on the allocation across specific products within it.

Note that the expenditure share of a product $j \in \mathcal{J}_{mt}^k$ can be expressed as

$$s_{jmt} = \frac{\exp\left(\frac{\delta_{jmt}}{1-\sigma}\right) \exp((1-\sigma)I_{mt}^k)}{\exp(I_{mt}^k) \exp(I_{mt})},$$

where $\delta_{jmt} = v_m^I - \Delta_m^k + \eta^k \mathbf{b}_{jmt} - \alpha \ln p_{jmt} + x'_{jmt} \beta + \lambda_t + \xi_{jmt}$ is the average utility, $I_{mt}^k \equiv \ln \sum_{j \in \mathcal{J}_{mt}^k} \exp\left(\frac{\delta_{jmt}}{1-\sigma}\right)$ and $I_{mt} \equiv \ln \sum_g \exp((1-\sigma)I_{mt}^g)$. Note that δ is defined inclusive of price, as opposed to the exposition in E.1. Define $\delta_{jmt} \equiv \bar{\delta}_{mt}^k + v_{jmt}$, which decomposes δ_{jmt} into a segment average and deviation from this average, where the first term is

$$\bar{\delta}_{mt}^k = \bar{v}_{mt}^k - \alpha \ln \bar{p}_{mt}^k + \beta' \bar{x}_{mt}^k + \lambda_t + \bar{\xi}_{mt}^k,$$

where, for any variable z , $\bar{z}_{mt}^k$ denotes the average of z_{jmt} across products in segment k . The second term can be written as

$$v_{jmt} = \dot{v}_{jmt}^k - \alpha \ln \dot{p}_{jmt}^k + \beta' \dot{x}_{jmt}^k + \dot{\xi}_{jmt}^k,$$

where $\dot{z}_{jmt}^k = z_{jmt} - \bar{z}_{mt}^k$ measure deviation of the value of z for a firm relative to the group average. We can then write

$$I_{mt}^k = \frac{\bar{\delta}_{mt}^k}{1-\sigma} + \ln N_{mt}^k + \ln \left(\frac{1}{N_{mt}^k} \sum_{j \in \mathcal{J}_{mt}^k} \exp\left(\frac{v_{jmt}}{1-\sigma}\right) \right),$$

where N_{mt}^k is the number of products in segment k and market m at t . Note that the last term is

zero if $v_{jmt} = 0 \quad \forall j \in \mathcal{J}_{mt}^k$, and that it is positive in any other case. We can then write

$$\begin{aligned} \ln N_{mt}^u s_{mt}^u &= \ln \sum_{j \in \mathcal{J}_{mt}^u} s_{jmt} \\ &= (1 - \sigma) I_{mt}^u - I_{mt} \\ &= \bar{\delta}_{mt}^u + (1 - \sigma) \ln N_{mt}^u + (1 - \sigma) \ln \left(\frac{1}{N_{mt}^u} \sum_{j \in \mathcal{J}_{mt}^u} \exp \left(\frac{v_{jmt}}{1 - \sigma} \right) \right) - I_{mt}. \end{aligned}$$

This expression leads to the following estimating equation for the unbranded segment

$$\ln s_{mt}^u - \ln s_{0mt} = \bar{v}_{mt}^u - \alpha \overline{\ln p}_{mt}^u + \beta' \bar{x}_{mt}^u + \lambda_t - \sigma \ln N_{mt}^u + v_{mt}^u,$$

with the term $-\sigma \ln N_{mt}^u$ reflecting the average conditional market share $1/N_{mt}^u$ among unbranded generics, and the composite error term

$$v_{mt}^u = \bar{\xi}_{mt}^u + (1 - \sigma) \ln \left(\frac{1}{N_{mt}^u} \sum_{j \in \mathcal{J}_{mt}^u} \exp \left(\frac{v_{jmt}}{1 - \sigma} \right) \right).$$

E.4 Compensating Variation

Denote $u_{ij} \equiv u(y_i, \delta_j, v_{ij})$, where market and time subscripts are omitted for notational convenience and $v_{ij} = \zeta_i^k + (1 - \sigma)\epsilon_{ij}$. The term δ_j includes the unobserved component ξ_j , which we hold fixed in our counterfactuals. Consider a policy that changes the initial $\delta = (\delta_1, \delta_2, \dots, \delta_J)$ to δ' , holding incomes and individual realizations of v fixed. The compensating variation for individual i , denoted CV_i , is defined by

$$\max_j u(y_i, \delta_j, v_{ij}) = \max_{j'} u(y_i - CV_i, \delta'_{j'}, v_{ij'}).$$

Thus, CV_i is the income reduction in the post-policy environment that leaves the consumer as well off as before the policy change. With the preferences given above, (log of) income is additively separable, and we can write

$$\begin{aligned} \alpha \varphi^{-1} [\ln y_i - \ln(y_i - CV_i)] &= \max_{j'} \{\delta'_{j'} + v_{ij'}\} - \max_j \{\delta_j + v_{ij}\} \equiv \Delta_i \\ CV_i &= y_i (1 - \exp(-\varphi \alpha^{-1} \Delta_i)). \end{aligned}$$

The expected compensating variation in the market can be obtained by integrating over y and

v . Assuming that y and v are independently distributed and thus $F_{y,v}(y, v) = F_y(y)F_v(v)$, we get

$$E[CV] = \int y dF_y(y) \int (1 - \exp(-\varphi\alpha^{-1}\Delta_i)) dF_v(v) = \bar{y} \int (1 - \exp(-\varphi\alpha^{-1}\Delta_i)) dF_v(v),$$

where $\bar{y}$ is the average income and the integral over the random utility components can be approximated using the procedure suggested by McFadden (1996), which can be simplified for this case as follows:

1. Draw a sequence of vectors v^t for $t = 1, \dots, T$, whose empirical distribution approximates F_v as $T \rightarrow \infty$.
2. For each draw, compute the maximum indirect utility for the baseline and counterfactual policy scenarios to obtain:

$$\Delta(v^t) \equiv \max_{j'} \{\delta'_{j'} + v^t_{j'}\} - \max_j \{\delta_j + v^t_j\}$$

3. Approximate $E[CV]$ as:

$$\overline{CV} = \bar{y} \frac{1}{T} \sum_{t=1}^T (1 - \exp(-\varphi\alpha^{-1}\Delta(v^t)))$$

For our analysis, we calibrate φ as the market size divided by the 2015 average monthly labor income as reported by the Chilean statistical office (\$722) multiplied by 12, and by the population size (17.97 million).

E.5 Supply Model Estimation Details

The parameters of interest on the supply side of the model are $\gamma_c, \gamma_F, \gamma_\kappa, \gamma_\pi$ and γ_{σ_e} , as discussed in Section 6.2. These parameters govern marginal costs, entry costs, certification costs, the distribution of drug quality, and profit shocks, respectively. In this appendix, we provide further details about the estimation procedure of these parameters. Throughout this procedure, demand is known to the econometrician.

The first step consists of recovering marginal costs by inverting the first-order condition in equation (8) as

$$\hat{c}_{jmt} = p_{jmt} + \frac{q_{jmt}(\mathbf{p}_{mt})}{\frac{\partial q_{jmt}(\mathbf{p}_{mt})}{\partial p_{jmt}}} \quad \forall j \in \mathcal{J}_{mt},$$

where the right-hand side depends on prices, drug attributes, and demand parameters, all of which are either observed or estimated at this point. To estimate γ_c , we use these fitted values for

marginal costs and estimate the following regression

$$\log \hat{c}_{jmt} = x'_{c,jmt} \gamma_c + \omega_{jmt},$$

where $x_{c,jmt}$ includes indicators for drug segments, markets, and years. This regression is estimated on the sample of drugs that sell in market m and year t . Assuming that all potential entrants share the same structure of marginal costs, that cost shocks ω_{jmt} are iid and only observed after entry, we can use these estimates to construct marginal costs and calculate variable profits for all potential entrants. We denote the estimated distribution of marginal costs as $\hat{G}_c$.

The second step recovers the remaining parameters relying on the entry and certification choices of manufacturers. We assume that profit shocks ε_{jmt}^1 and ε_{jmt}^0 are iid T1EV. Given this distributional assumption, the entry probabilities in equations (6) and (7) have the usual logit form. In particular, the entry probabilities for environments without and with quality regulation NQ and Q , respectively, are

$$\begin{aligned} \phi_{jmt}^{NQ} &= \frac{\exp(\frac{1}{\sigma_\varepsilon} [E_{\hat{\phi}_{jmt}^{NQ}}[\tilde{\Pi}_{jmt}(\mathcal{J})] - x'_{F,jm} \gamma_F])}{1 + \exp(\frac{1}{\sigma_\varepsilon} [E_{\hat{\phi}_{jmt}^{NQ}}[\tilde{\Pi}_{jmt}(\mathcal{J})] - x'_{F,jm} \gamma_F])} \\ \phi_{jmt}^Q &= \frac{\exp(\frac{1}{\sigma_\varepsilon} [E_{\hat{\phi}_{jmt}^Q}[\tilde{\Pi}_{jmt}(\mathcal{J})] - x'_{F,jm} \gamma_F - (1 - d_j^l) x'_{\kappa,jm} \gamma_\kappa])}{1 + \exp(\frac{1}{\sigma_\varepsilon} [E_{\hat{\phi}_{jmt}^Q}[\tilde{\Pi}_{jmt}(\mathcal{J})] - x'_{F,jm} \gamma_F - (1 - d_j^l) x'_{\kappa,jm} \gamma_\kappa])} \left[\frac{\exp(x'_{\pi,jm} \gamma_\pi)}{1 + \exp(x'_{\pi,jm} \gamma_\pi)} \right]^{1-d_j^l}, \end{aligned}$$

where d_j^l indicates that j is an innovator drug.

We then use the probabilities to estimate the remaining model parameters by maximum likelihood. The log-likelihood of the observed market structure for environments without and with quality regulation, respectively, are

$$\begin{aligned} \log \ell_{jmt}^{NQ} &= y_{jmt} \log \phi_{jmt}^{NQ} + (1 - y_{jmt}) \log(1 - \phi_{jmt}^{NQ}) \\ \log \ell_{jmt}^Q &= y_{jmt} \log \phi_{jmt}^Q + (1 - y_{jmt}) \log(1 - \phi_{jmt}^Q), \end{aligned} \tag{A.6}$$

To compute the likelihood, we need several inputs. The first input are variable profits $\tilde{\Pi}_{jmt}(\mathcal{J})$ for any potential market structure $\mathcal{J}$. For a given $\mathcal{J}$ and draws of marginal costs from $\hat{G}_c$, it is straightforward to compute variable profits by solving the second-stage pricing game. Taking the expectation over variable profits requires integrating over the distribution of market structures since this is a private information game. However, these probabilities are an endogenous outcome in the model, as discussed in Section 5.2. To deal with this issue, Seim (2006) develops a nested likelihood estimator that exploits the equilibrium entry conditions from the model. This would be extremely computationally costly given the number of markets and potential market structures

in our data. Instead, we follow Sweeting (2009) and compute these probabilities directly from the data. In particular, we estimate logit regressions of entry and certification choices on all their determinants as predicted by the model, namely own and rival drivers of variable profits, fixed costs, and certification costs. This is

$$P(y_{jmt} = 1) = \Lambda(x'_{jmt}\eta),$$

where y_{jmt} is an indicator for entry of firm j in market m in year t , x_{jmt} is a vector of firm and market characteristics that includes segment dummies, the segment shares of rival potential entrants per segment, a dummy for being previously in the market, the number of rival firms previously in the market, market fixed effects, and log market size. We estimate this model using data from the first and last years of our sample, 2010 and 2017, to approximate environments without and with quality regulation, respectively. We use the fitted values from this regression as estimates of firm entry probabilities, $\hat{\phi}_{jmt}^{NQ}$ and $\hat{\phi}_{jmt}^Q$.

The number of potential market structures is remarkably large at 2^{P_m} , where P_m is the number of potential entrants in a given market. Having entry probabilities, we take $N_{MS} = 200$ Halton draws of market structures. Denoting each drawn market structure $\mathcal{J}_{mt}^{(s),NQ}$ and $\mathcal{J}_{mt}^{(s),Q}$ for environments, we approximate expected variable profits as

$$\begin{aligned} E_{\hat{\phi}^{NQ}}[\tilde{\Pi}_{jmt}(\mathcal{J})] &\approx \frac{1}{N_{MS}} \sum_{s=1}^{N_{MS}} \tilde{\Pi}_{jmt}(\mathcal{J}_{mt}^{(s),NQ}) \\ E_{\hat{\phi}^Q}[\tilde{\Pi}_{jmt}(\mathcal{J})] &\approx \frac{1}{N_{MS}} \sum_{s=1}^{N_{MS}} \tilde{\Pi}_{jmt}(\mathcal{J}_{mt}^{(s),Q}). \end{aligned}$$

Having expected variable profits, estimation proceeds directly by maximum likelihood, following equation (A.6). We obtain standard errors following the standard procedure based on the inverse of the Hessian matrix.

There are a few practical details in this estimation procedure that are worth mentioning. First, we must take a stance on the set of potential entrants in the entry model. For each market, we include all the firms that ever participate in the market during our sample period, along with the five firms with the highest participation in each segment across markets at the national level. Second, we use the conditional mean of marginal costs based on $\hat{G}_c$ when computing expected variable profits to reduce computational burden. Third, we impose that potential entrants enter with a single presentation and draw their bioequivalent status from the empirical distribution of this variable. Fourth, in terms of the estimating sample, we focus on the first and last year in our data. By focusing on those two cross sections, we aim at letting our estimates be mostly informed by environments without and with quality regulation rather than by the intermediate transition

period. Finally, to ease interpretation, we use an annual discount rate of 0.05 to take the present value of a stream of variable profits. By doing this, we put variable profits in the same units as fixed and certification costs.

E.6 Model Simulation Details

To implement our counterfactual analysis in Section 7, we simulate equilibrium outcomes using our estimated model. We start from a dataset that includes all potential entrants to a market, their segments, and product attributes, along with estimates of demand, marginal costs, entry costs, certification costs, the distribution of drug quality, and the scale of profit shocks. For each market in this dataset, we proceed as follows:

1. To start simulation s , take $N_{MS} = 50$ draws of potential market structures from the distribution of market structures.
2. Take a draw of profit shocks $(\tilde{\varepsilon}_{0j}^{(s)}, \tilde{\varepsilon}_{1j}^{(s)})$ for each potential entrant. Take a draw of quality $\tilde{\psi}_j^{(s)}$ for each potential entrant.
3. For each market structure simulated in step 1, solve for optimal pricing and compute variable profits for all entrants using our demand and marginal cost estimates. Compute expected variable profits for each potential entrant across all draws of potential market structures.
4. Using expected variable profits from step 3, estimates of fixed costs and certification costs (when considering environments with quality regulation), draws of drug quality $\tilde{\psi}_j^{(s)}$ from step 2, and an initial guess for entry probabilities as inputs, solve for a fixed point in entry probabilities.
5. Using equilibrium entry probabilities from step 4 and draws of profit shocks $(\tilde{\varepsilon}_{0j}^{(s)}, \tilde{\varepsilon}_{1j}^{(s)})$ from step 2, compute realized equilibrium market structure, solve for optimal price, and compute demand given that market structure.
6. Repeat this procedure from 1 to 5 for $s = 1, \dots, N_S$ runs, so we can then average across simulations to smooth the realizations of shocks. The results in the paper average across $N_S = 50$ runs of this procedure.

To obtain the results in the paper, we use this procedure to generate a baseline market structure over which to develop our counterfactual analysis. We proceed as follows:

1. We start from an environment without quality regulation to simulate a baseline market structure. This market structure reflects the fact that when regulation is imposed, a particular market structure is already in place, generated by the same underlying parameters for demand, marginal costs, quality, and entry costs.

2. Taking the baseline market structure as given and known to all firms, we simulate market outcomes again under different environments ranging from no regulation to particular forms of regulation. Depending on the counterfactual, we adjust various parameters of the model for each simulation. The results we report in the paper correspond to equilibrium outcomes from this second simulation.

E.7 Robustness of Welfare Analysis

The welfare effects of the reform depend on the value of low-quality generics vis-a-vis high-quality generics. In our baseline welfare calculations, we normalize this value to zero. Here, we discuss the need for a normalization and the robustness of our estimates regarding the welfare effects of the reform under alternative assumptions.

A more general version of the utility function we use in the main text can be written as

$$v_{jmt}^k = \mu_m^I \cdot \left(E[\psi_j | k, \mathbf{b}_{jmt}] + (1 - E[\psi_j | k, \mathbf{b}_{jmt}]) \cdot \frac{\mu_m^k}{\mu_m^I} - \tau_m^k(b_{jmt}) \right).$$

This expression differs from equation (4) in the paper by including the term $(1 - E[\psi_j | k, \mathbf{b}_{jmt}]) \cdot \frac{\mu_m^k}{\mu_m^I}$, which reflects that non-bioequivalent drugs could in principle have a value μ_m^k different than zero. In this case, and adopting the notation in Section 6.1, the relationship between the structural parameters and the estimated parameters is

$$\begin{aligned} v_m^I &= \mu_m^I \\ \Delta_m^k &= \mu_m^I(1 - \pi_H^k + \tau_{0,m}^k) - \mu_m^k(1 - \pi_H^k) \\ \eta^k &= \mu_m^I(1 - \pi_H^k + \tau_{0,m}^k - \tau_{1,m}^k) - \mu_m^k(1 - \pi_H^k). \end{aligned}$$

These expressions highlight the need for a normalization, because there are three equations and four unknown parameters. Intuitively, it is not possible to separately identify the value of generics and the level of bias against them. In the text, we normalize the value of non-bioequivalent generics μ_m^k to 0, and the previous expressions simplify to those in equation (10). This normalization is consequential for our estimates of the welfare effects of quality regulation. The lower the value of μ_m^k , the higher the welfare losses from purchasing low-quality drugs in the pre-reform period and, therefore, the higher welfare gains from the reform.

To analyze the robustness of our welfare results parsimoniously, we consider cases in which

$$\mu_m^k = \rho \mu_m^I \quad \text{for} \quad \rho \in [0, 1), \quad (\text{A.7})$$

such that we set the value of non-bioequivalent generics equal to a fraction ρ of the value of the

innovator and recompute our welfare results for different levels of ρ . The value of ρ parameterizes the medical value of bioequivalence. The case of $\rho = 0$ is the benchmark case in the main text and assumes there is no medical value in consuming a drug that is not therapeutically equivalent to the innovator. Conversely, the case of $\rho = 1$ corresponds to assuming that therapeutical equivalence is inconsequential for welfare; i.e., that non-bioequivalent generics provide the same medical value to patients as the innovator. This is, evidently, an inconvenient assumption, as it would not be possible to justify the widespread adoption of quality regulation based on bioequivalence in this case. The value of ρ only affects the calculation of consumer welfare. The effects of the reform on market shares, prices, market structure, and profits depend only on how *choice utility* changes with the reform and on the supply-side parameters, which are invariant to ρ .

Although we believe that the case of $\rho = 0$ is a natural benchmark, it is ultimately an untestable assumption, and it could be the case that non-bioequivalent drugs provide *some* value to patients. Figure A.6 shows the welfare results for different assumptions on ρ . The figure shows the impacts of quality regulation on consumer welfare and total welfare under both assumptions regarding the part of the aversion to generics that is welfare-relevant (assumptions A1 and A2). As discussed above, the welfare effect of the reform decreases with ρ . However, even for very high values of ρ , the welfare effects of the reform are positive. In other words, the reform produces welfare gains even with small welfare losses from buying a non-bioequivalent drug.

References

- CHANDRA, A., FLACK, E. and OBERMEYER, Z. (2024). The Health Costs of Cost Sharing. *The Quarterly Journal of Economics*, **139** (4), 2037–2082.
- DEIS (2019). Egresos Hospitalarios. <https://deis.minsal.cl/#datosabiertos>.
- DUBOIS, P. and LASIO, L. (2018). Identifying Industry Margins with Price Constraints: Structural Estimation on Pharmaceuticals. *American Economic Review*, **108** (12), 3685–3724.
- FOSTER, L., HALTIWANGER, J. C. and KRIZAN, C. J. (2001). Aggregate Productivity Growth: Lessons from Microeconomic Evidence. In *New Developments in Productivity Analysis*, University of Chicago Press, pp. 303–372.
- HUANG, D. and ROJAS, C. (2013). The Outside Good Bias in Logit Models of Demand with Aggregate Data. *Economics Bulletin*, **33** (1), 198–206.
- and — (2014). Eliminating the Outside Good Bias in Logit Models of Demand with Aggregate Data. *Review of Marketing Science*, **12** (1), 1–36.
- HUBER, C. A., SZUCS, T. D., RAPOLD, R. and REICH, O. (2013). Identifying Patients with Chronic Conditions using Pharmacy Data in Switzerland: An Updated Mapping Approach to the Classification of Medications. *BMC Public Health*, **13**, 1–10.
- INE (2018). VIII Encuesta de Presupuestos Familiares (EPF), 2016–2017. <https://www.ine.gob.cl/estadisticas/sociales/ingresos-y-gastos/encuesta-de-presupuestos-familiares>, accessed: 2026-09-18.

- JIN, G. Z. and LESLIE, P. (2003). The Effect of Information on Product Quality: Evidence from Restaurant Hygiene Grade Cards. *The Quarterly Journal of Economics*, **118** (2), 409–451.
- McFADDEN, D. (1996). Computing Willingness-to-Pay in Random Utility Models. *University of California at Berkeley, Econometrics Laboratory Software Archive, Working Papers*.
- NEVO, A. and HATZITASKOS, K. (2006). *Why Does the Average Price Paid Fall during High Demand Periods?* Tech. rep., CSIO working paper.
- SEIM, K. (2006). An Empirical Model of Firm Entry with Endogenous Product-Type Choices. *The RAND Journal of Economics*, **37** (3), 619–640.
- SWEETING, A. (2009). The Strategic Timing Incentives of Commercial Radio Stations: An Empirical Analysis Using Multiple Equilibria. *The RAND Journal of Economics*, **40** (4), 710–742.
- WHO (2007). Appendix C: Classification of Drugs by American Hospital Formulary Services List Number and their ICD-9-CM Equivalents. https://theodora.com/diseases/appendix_c.html.

Figure A.1: Labeling of bioequivalent drugs



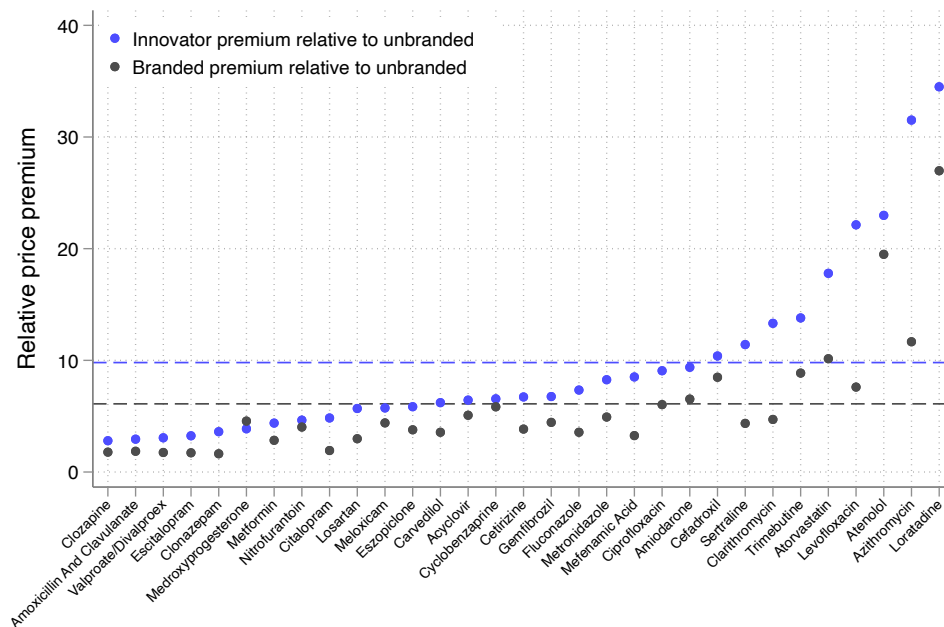
(a) Instructions for bioequivalent drugs labeling



(b) Examples of labeled bioequivalent drugs

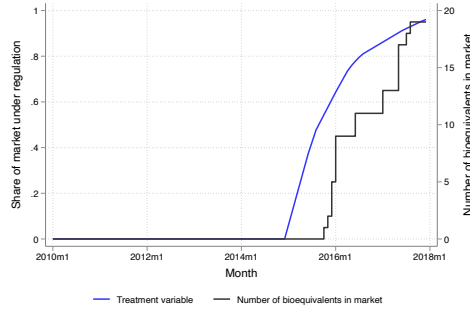
Notes: These figures display both instructions and examples of required labeling of bioequivalent drugs. The objective of this labeling was to highlight drugs with bioequivalence approval.

Figure A.2: Innovator and branded generics price premiums relative to unbranded generics

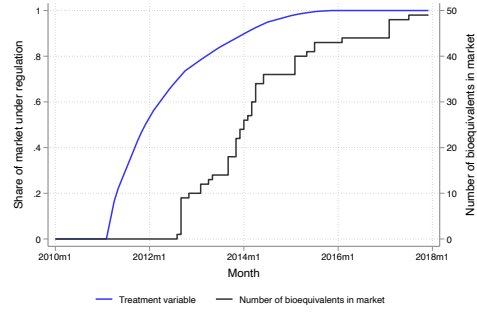


Notes: This figure displays the estimated price premium for innovator and branded generic drugs relative to unbranded generic drugs. Each dot in the figure corresponds to an exponentiated coefficient from a regression of log prices on innovator and branded drug dummies, estimated separately for each molecule using data for 2010-2011. The sample includes all markets with price information for at least one innovator, one branded, and one unbranded drug during that period. We exclude one molecule with an outlier estimated premium. Dashed lines indicate the average price premium across this set of molecules.

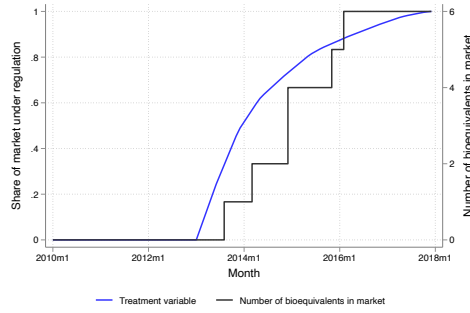
Figure A.3: Policy variation induced by bioequivalence requirements



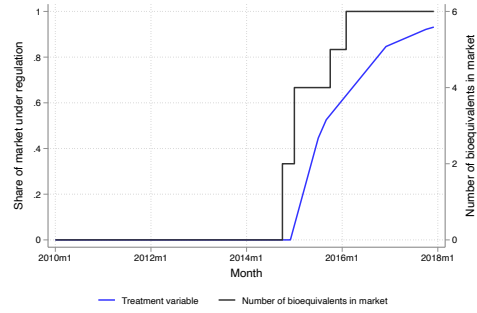
(a) Aripiprazole



(b) Atorvastatin



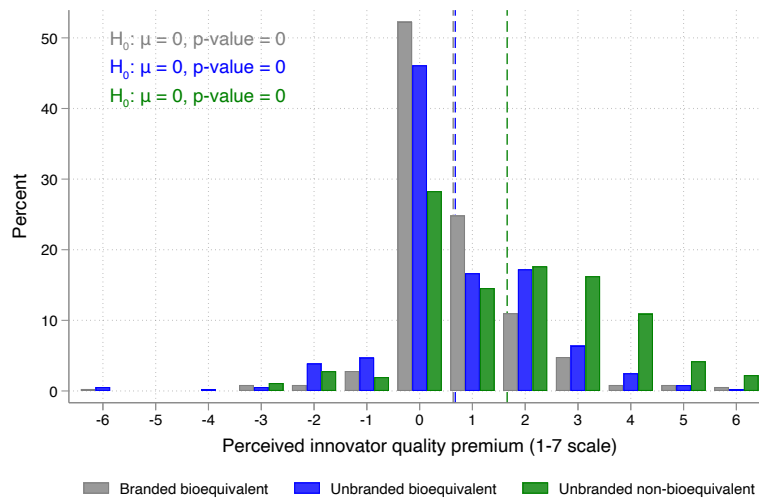
(c) Citalopram



(d) Deflazacort

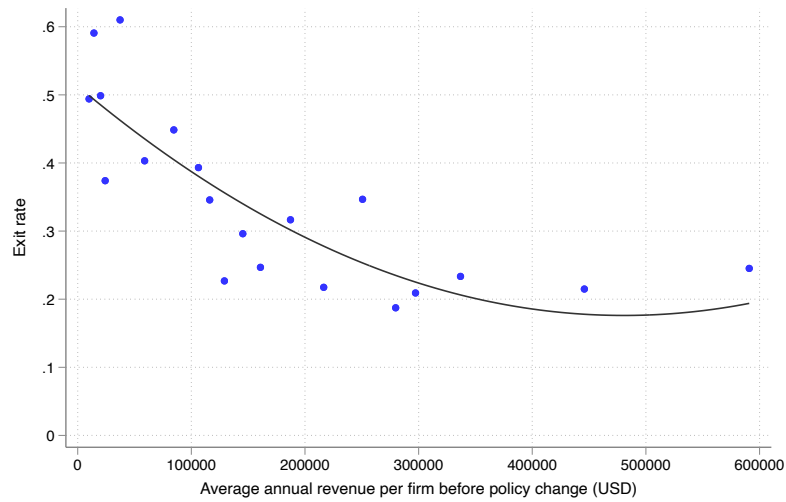
Notes: Each figure displays the values of the treatment variable and the number of bioequivalent drugs in a different market. This version of the treatment variable uses the first deadline as the relevant date. The treatment variable is displayed in blue and takes a value of 0 before the first deadline, and then increases as renewal dates of drugs in the molecule approach. The number of bioequivalent drugs in the molecule is displayed in gray. These four examples are plotted along all other markets in our sample in Figure 2-c.

Figure A.4: Survey results for perceived quality



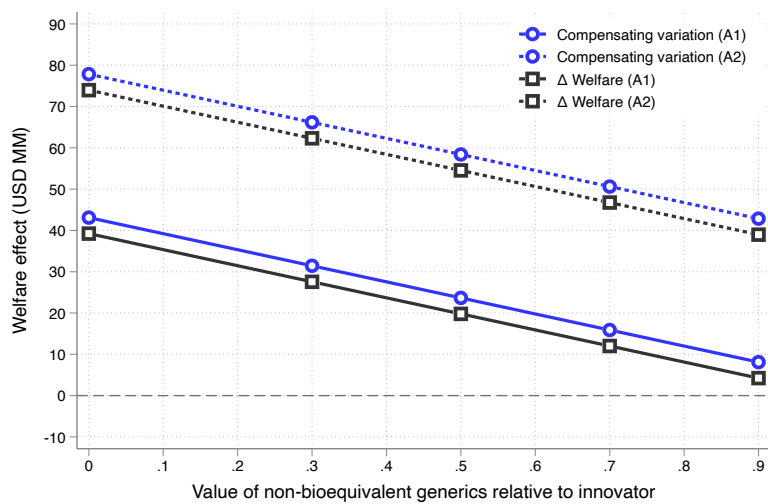
Notes: This figure displays the distribution of perceived quality premiums for different drug segments relative to the innovator drug, namely a branded bioequivalent generic (gray), an unbranded bioequivalent generic (blue), and an unbranded non-bioequivalent generic (green). The premium is calculated as the difference between the perceived quality of the innovator drug and the perceived quality for each drug segment, which is recorded in a 1-7 scale. Dashed lines indicate the average for each drug segment in the figure.

Figure A.5: Correlation between drug exit after regulation and baseline revenue



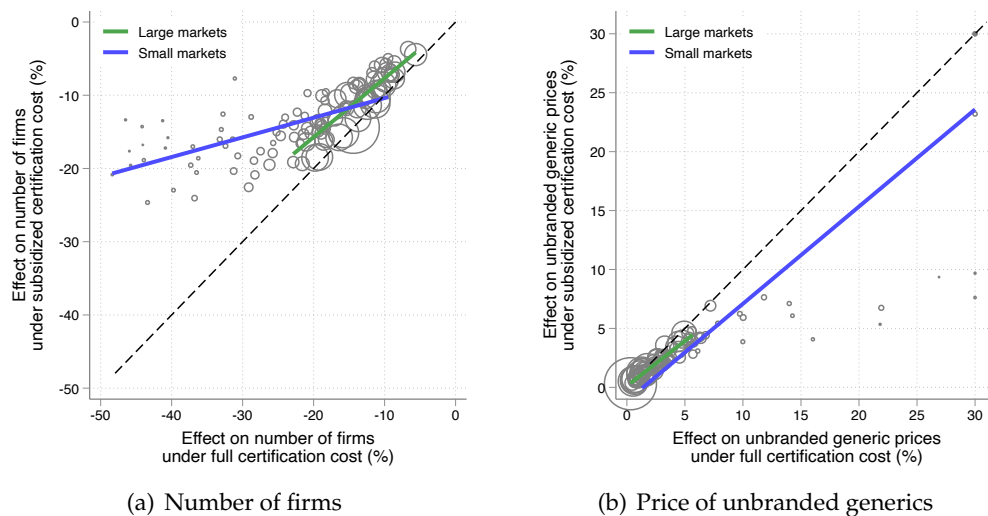
Notes: This figure shows a binned scatter plot for the relationship between exit rate after the policy change and average annual revenue before the policy change. The black line indicates a quadratic fit of this relationship.

Figure A.6: Welfare effects under alternative assumptions for non-bioequivalent drugs' valuation



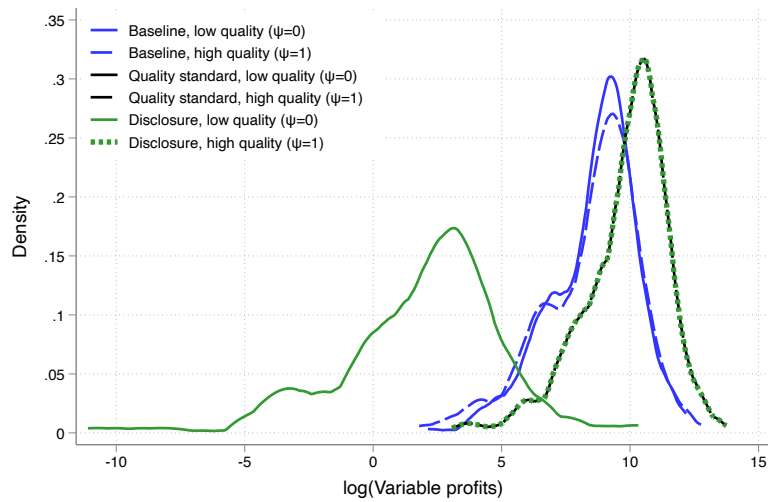
Notes: This figure shows the impact of quality regulation on consumer welfare (blue) and total welfare (black) for different assumptions regarding the ratio between the value of non-bioequivalent generics and the value of innovators, denoted by ρ in equation (A.7). The results for $\rho = 0$ correspond to the benchmark case we focus on throughout the paper. We report results for welfare assumptions A1 (solid) and A2 (dashed).

Figure A.7: Heterogeneity in the effects of quality regulation and the role of certification costs



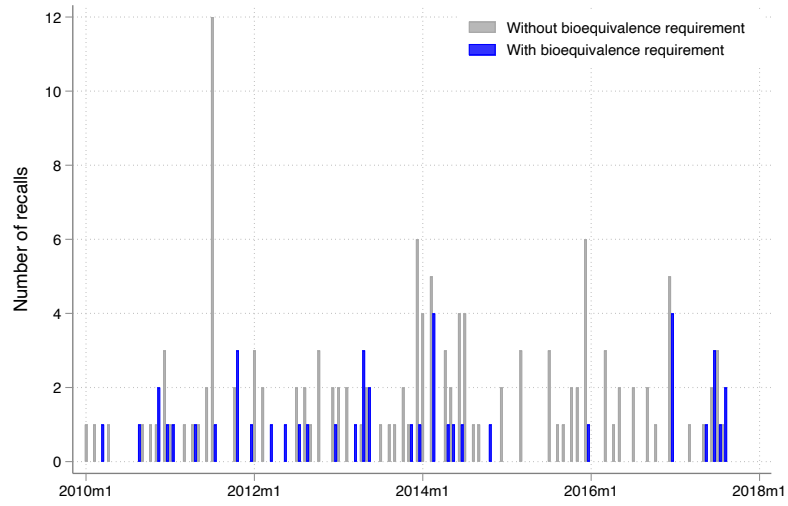
Notes: These figures compare simulated effects of quality regulation across environments where firms must pay the full certification cost (x-axis), and where the certification cost is fully subsidized (y-axis). Each circle is a market, and the size of each circle indicates market size. The green (blue) line is a linear fit for large (small) markets, defined around the median market size. The dashed line is a 45-degree line. Panel (a) displays results for the effect of quality regulation on the number of firms in the market, Panel (b) displays results for the effect of quality regulation on the average price of unbranded generics in the market.

Figure A.8: The impact of a quality standard and quality disclosure on variable profits



Notes: This figure displays the density of log variable profits among firms that enter the market in three environments: a baseline without regulation (blue), a quality standard (black), and quality disclosure (green). Solid lines indicate variable profits for low-quality drugs, whereas dashed lines indicate profits for high-quality firms.

Figure A.9: Number of recalls per month



Notes: The figure shows the number of product recalls over time split into markets with bioequivalence requirements and markets without bioequivalence requirements.

Figure A.10: Consumer survey elicitation of perceived quality and price

4 variedades de Atorvastatina para el Colesterol, todas con la misma dosis y número de tabletas



Lipitor - Laboratorio Pfizer
Medicamento Original



Atorvastatina - Laboratorio Mintlab
Genérico sin Marca - Bioequivalente



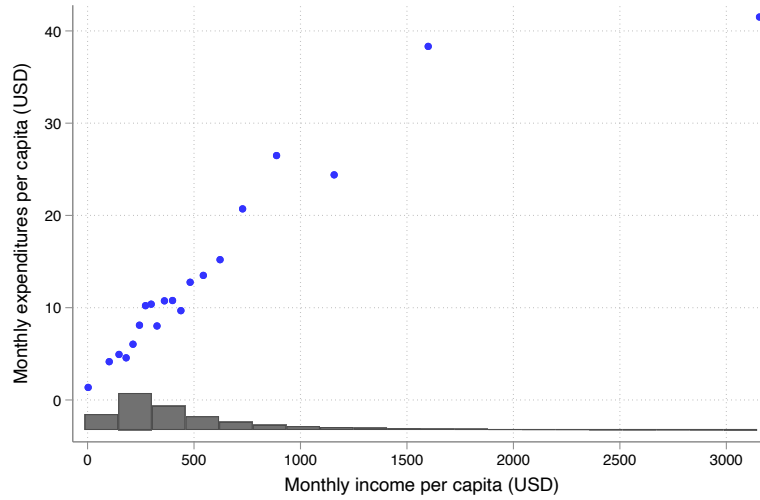
Atorvastatina - Laboratorio Mintlab
Genérico sin Marca - No Bioequivalente



Lipoten - Laboratorio Pharmavita
Medicamento de Marca - Bioequivalente¹

Notes: This figure displays the sheet surveyors provided consumers in our survey sample. This sheet displays the 4 drugs we used as an example to elicit perceived quality and price differences. While observing this sheet, surveyors asked consumers first to assign a score in a 1-7 scale to each drug regarding their quality, and then to estimate the price of each drug given that the innovator had a price of \$50,000 CLP (\$71.43 USD).

Figure A.11: Drug expenditures and income



Notes: This figure displays a binned scatterplot of monthly income per capita and monthly drug expenditures per capita at the household level using data from the 2016 National Household Spending Survey (INE, 2018). The graph also includes a histogram of monthly per-capita household income.

Table A.1: Timing of reform to quality regulation

		<i>A - Relevant policy dates</i>				<i>B - Market characteristics</i>					
Group	Number of molecules	First decree		Last decree		Number of drugs	Average price	Average revenue	Share of drugs by segment		
		Decree	Deadline	Decree	Deadline				Innovator	Branded	Unbranded
1	3	2011-01	2011-02	2013-06	2013-12	60	1.58	20,065	0.18	0.74	0.08
2	19	2011-01	2012-02	2013-06	2013-12	158	3.86	19,774	0.27	0.62	0.11
3	7	2012-10	2013-10	2013-10	2014-04	54	0.68	13,167	0.13	0.67	0.20
4	24	2012-12	2013-12	2012-12	2013-12	264	4.39	12,839	0.21	0.69	0.10
5	20	2012-12	2013-01	2013-06	2013-12	282	2.43	15,840	0.18	0.77	0.05
6	9	2012-12	2014-12	2015-12	2016-12	101	4.86	13,660	0.17	0.79	0.05
7	15	2012-12	2014-12	2016-12	2017-06	210	4.14	13,438	0.19	0.76	0.05
8	10	2012-12	2014-12	2016-12	2017-12	82	2.34	13,369	0.18	0.77	0.05
9	8	2014-02	2015-12	2016-12	2017-12	18	3.46	4,960	0.00	0.41	0.59

Notes: Panel A in this table displays the dates of announcement and deadlines of BE requirements for different groups of molecules included in our sample. The groups are defined as a unique combination of decrees and deadlines. Panel B in this table displays average product characteristics in 2010, by molecule groups. Prices per defined daily dose and revenues are measured in 2017 USD.

Table A.2: Descriptive statistics

Variable	N	Mean	SD	p10	p50	p90
<i>A - Price per DDD</i>						
Innovators	24,808	6.19	8.11	0.80	3.44	14.64
Branded generics	79,141	2.67	3.49	0.39	1.60	5.90
Unbranded generics	9,792	0.77	1.53	0.04	0.24	1.98
Bioequivalents	16,997	2.51	3.39	0.36	1.49	5.50
<i>B - Market shares</i>						
Innovators	11,040	0.29	0.29	0.00	0.23	0.75
Branded generics	11,040	0.44	0.33	0.00	0.44	0.88
Unbranded generics	11,040	0.27	0.37	0.00	0.00	1.00
Bioequivalents	11,040	0.09	0.19	0.00	0.00	0.37
<i>C - Number of drugs</i>						
Innovators	11,040	2.34	2.17	0.00	2.00	5.00
Branded generics	11,040	12.78	13.18	1.00	9.00	29.00
Unbranded generics	11,040	6.74	6.34	1.00	5.00	16.00
Bioequivalents	11,040	2.85	6.49	0.00	0.00	9.00
<i>D - Number of laboratories</i>						
Innovators	11,040	0.88	0.42	0.00	1.00	1.00
Branded generics	11,040	6.52	4.48	1.00	6.00	12.00
Unbranded generics	11,040	4.28	3.30	1.00	4.00	9.00
Bioequivalents	11,040	1.78	3.49	0.00	0.00	6.00

Notes: This table displays descriptive statistics from the IQVIA data and the ISP registry data. Statistics for prices come from IQVIA and are displayed in 2017 USD and calculated at the drug level, while the remainder are calculated at the market level. Statistics for market shares come from IQVIA and are only observed for markets in which at least one drug is sold in the period. Statistics for the number of drugs and laboratories come from the ISP registry, are computed using only observations with a valid marketing license.

Table A.3: First stage of demand estimation

	(1)	(2)	(3)	(4)
	Dependent variable			
	$\ln s_{jmt k}$	$\ln p_{jmt}$	b_{jmt}^B	b_{jmt}^U
Average age of competitors	-1.95 (0.22)	-0.20 (0.03)	0.05 (0.03)	-0.01 (0.00)
log Price, Norway	0.01 (0.16)	0.19 (0.04)	-0.16 (0.02)	0.02 (0.00)
Past renewal, branded	-0.27 (0.19)	-0.07 (0.03)	0.37 (0.03)	-0.01 (0.00)
Past renewal, unbranded	-0.64 (0.26)	0.20 (0.10)	-0.67 (0.04)	1.08 (0.07)
Observations	4,011	4,011	4,011	4,011
R^2	0.44	0.92	0.51	0.77
F-stat IV	19.61	17.45	234.77	68.02

Notes: This table displays first-stage regressions of each endogenous variable in the model on the instrumental variables. The specification includes market and year fixed effects. Standard errors in parentheses.

Table A.4: Effects of quality regulation on proxies of drug quality

	(1)	(2)	(3)	(4)
	Drug adverse effects			Drug
	Admissions	Hospital days	Surgeries	recalls
<i>A - Average effects</i>				
Regulation	-0.02 (0.03)	-0.13 (0.12)	0.00 (0.00)	0.00 (0.00)
R^2	0.81	0.85	0.17	0.21
<i>B - Heterogeneity by market size</i>				
Regulation $\times$ Low revenue	-0.08 (0.09)	-0.26 (0.25)	0.00 (0.00)	0.00 (0.00)
Regulation $\times$ High revenue	0.03 (0.04)	-0.00 (0.01)	0.00 (0.00)	-0.00 (0.00)
R^2	0.81	0.85	0.17	0.21
Pre-regulation average	0.09	0.17	0.00	0.00
Observations	568	568	568	870
Market FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y

Notes: Each column in this table shows the estimated coefficients from a regression of an outcome related to drug quality on the policy roll-out variable constructed using the first decree deadline, as in equation (A.2). Outcomes are constructed as the ratio of the variable of interest over drug sales measured in thousands of daily doses. Columns 1-3 are related to adverse health effects, whereas column 4 is related to drug recalls, and includes all recalls. Panel B provides results by baseline revenue. Markets are classified as having low or high revenue based on the total market revenue in 2010 relative to the median across markets in that year. Clustered standard errors are reported in parentheses. ***p<0.01, **p<0.05, *p<0.1.

Table A.5: Summary statistics from consumer survey data

Variable	N	Mean	SD	p10	p50	p90
Perceived quality of innovator drug (1-7)	361	6.31	1.01	5.00	7.00	7.00
Perceived quality of bioequivalent branded drug (1-7)	378	5.68	1.31	4.00	6.00	7.00
Perceived quality of bioequivalent unbranded drug (1-7)	386	5.62	1.29	4.00	6.00	7.00
Perceived quality of non-bioequivalent unbranded drug (1-7)	381	4.67	1.66	3.00	5.00	7.00
Perceived price of bioequivalent branded drug (USD)	398	36.24	20.19	8.57	35.71	64.29
Perceived price of bioequivalent unbranded drug (USD)	401	22.41	15.68	4.29	21.43	42.86
Perceived price of non-bioequivalent unbranded drug (USD)	399	18.01	14.24	2.86	14.29	35.71
Recognizes bioequivalent drug label	401	0.84	0.37	0.00	1.00	1.00
Understanding about bioequivalence (1-5)	401	2.91	1.47	1.00	3.00	5.00
=1 if doctors specify brand in prescriptions	299	0.65	0.48	0.00	1.00	1.00
=1 if always purchases doctor recommendation	310	0.15	0.36	0.00	0.00	1.00
=1 if sometimes deviate from doctor recommendation	310	0.52	0.50	0.00	1.00	1.00
=1 if always chooses cheapest available drug	310	0.34	0.47	0.00	0.00	1.00
Purchases innovator drugs	338	0.41	0.49	0.00	0.00	1.00
Purchases bioequivalent branded drugs	338	0.20	0.40	0.00	0.00	1.00
Purchases bioequivalent unbranded drugs	338	0.28	0.45	0.00	0.00	1.00
Purchases non-bioequivalent unbranded drugs	338	0.11	0.31	0.00	0.00	1.00
Chronic illness by household member	401	0.58	0.49	0.00	1.00	1.00
Atorvastatin consumption by household member	401	0.34	0.48	0.00	0.00	1.00

Notes: This table displays summary statistics from our consumer survey. The total number of surveys is $N = 401$. Whenever the number of observations is smaller, it is due to the consumer not answering the question, except for the case of questions regarding physician prescription behavior, which were added to the survey with a lag and are therefore not available for around a fourth of the sample.

Table A.6: Decomposition of the effects of quality regulation on drug prices

	(1)	(2)	(3)	(4)
	All drugs	Innovator	Branded generics	Unbranded generics
Dep. var.: Contribution of changes in prices ($\hat{P}_{PC}$)	0.07** (0.03)	0.01 (0.03)	0.00 (0.02)	0.15*** (0.06)
R^2	0.63	0.70	0.67	0.65
Dep. var.: Contribution of changes in shares ($\hat{P}_{RW}$)	0.03 (0.04)	0.05* (0.03)	-0.00 (0.02)	0.01 (0.01)
R^2	0.53	0.65	0.53	0.49
Dep. var.: Contribution of correlation between shares and prices ($\hat{P}_{CS}$)	0.00 (0.01)	-0.01 (0.02)	-0.05 (0.03)	0.01 (0.01)
R^2	0.49	0.63	0.48	0.29
Dep. var.: Contribution of drug entry ($\hat{P}_E$)	0.03** (0.02)	0.00 (0.01)	0.04** (0.02)	-0.01 (0.01)
R^2	0.52	0.61	0.61	0.44
Dep. var.: Contribution of drug exit ($\hat{P}_X$)	-0.01* (0.00)	-0.02** (0.01)	-0.00 (0.01)	0.00 (0.00)
R^2	0.38	0.28	0.31	0.06
Observations	11,040	8,141	8,707	5,238
Market FE	Y	Y	Y	Y
Month-Segment FE	Y	Y	Y	Y

Notes: The table displays results for each component of the decomposition of price changes in equation (A.1). Each coefficient comes from a separate regression of the component indicated in the left for the drug segment indicated in the top row on the policy roll-out variable constructed using the first decree deadline. The sample size in column 1 is the number of markets (115) by the number of months (96). The sample size in columns 2–4 corresponds to the number of market-month observations with at least one product in the segment. We also drop markets without products in the segment at baseline (January 2010). Standard errors clustered at the market level are reported in parentheses. ***p<0.01, **p<0.05, *p<0.1.

Table A.7: Effects of quality regulation on number of drugs by drug attributes

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Dep. var.: Number of drugs							
	All	Innovator	Branded generics			Unbranded generics		
			All	BE	Non-BE	All	BE	Non-BE
<i>A - Heterogeneity by chronic condition</i>								
Regulation × Non-chronic	-0.26*** (0.07)	0.01 (0.08)	-0.28*** (0.07)	2.12*** (0.25)	-0.41*** (0.08)	-0.26*** (0.08)	2.64*** (0.42)	-0.45*** (0.08)
Regulation × Chronic	-0.11** (0.05)	-0.08 (0.07)	-0.10* (0.06)	1.31*** (0.28)	-0.32*** (0.07)	-0.11 (0.09)	2.90*** (0.55)	-0.26*** (0.08)
<i>B - Heterogeneity by mortality effects</i>								
Regulation × Low mortality effect	-0.20*** (0.05)	-0.03 (0.06)	-0.21*** (0.05)	1.52*** (0.27)	-0.38*** (0.07)	-0.19*** (0.07)	2.83*** (0.43)	-0.31*** (0.08)
Regulation × High mortality effect	-0.20** (0.10)	0.02 (0.11)	-0.19* (0.10)	1.97*** (0.28)	-0.37** (0.15)	-0.21** (0.10)	2.51*** (0.55)	-0.52*** (0.11)
Pre-regulation average	22.06	2.40	12.32	0.09	12.23	7.34	0.01	7.33
Observations	11,040	9,600	10,272	8,064	10,272	10,560	7,008	10,560
Market FE	Y	Y	Y	Y	Y	Y	Y	Y
Month FE	Y	Y	Y	Y	Y	Y	Y	Y

Notes: Each column in this table is a Poisson regression for the number of drugs in a segment on the policy roll-out variable constructed using the first decree deadline. Panel A provides heterogeneity results by whether the drug treats a chronic condition, following the classification in Huber *et al.* (2013). Panel B provides heterogeneity results by whether the drug has high mortality effects, following the classification in Chandra *et al.* (2024). The sample size in column 1 is the number of markets (115) by the number of months (96). Columns 2–8 omit markets with no variation in the number of products in the segment. Standard errors clustered at the market level are reported in parentheses. ***p<0.01, **p<0.05, *p<0.1.

Table A.8: Effects of quality regulation on drug prices by drug attributes

	(1)	(2)	(3)	(4)
	Dep. var.: Drug Price Index ($\hat{P}_{mt}$)			
	All drugs	Innovator	Branded generics	Unbranded generics
<i>A - Heterogeneity by chronic condition</i>				
Regulation $\times$ Non-chronic	0.15** (0.06)	0.06* (0.03)	-0.02 (0.05)	0.23*** (0.07)
Regulation $\times$ Chronic	0.08 (0.08)	0.01 (0.04)	0.01 (0.05)	0.07 (0.07)
R^2	0.99	0.99	0.99	0.99
<i>B - Heterogeneity by mortality effects</i>				
Regulation $\times$ Low mortality effect	0.10* (0.06)	0.02 (0.03)	-0.04 (0.05)	0.14** (0.06)
Regulation $\times$ High mortality effect	0.23** (0.09)	0.15*** (0.05)	0.16* (0.09)	0.23** (0.11)
R^2	0.99	0.99	0.99	0.98
Observations	11,040	8,141	8,707	5,238
Market FE	Y	Y	Y	Y
Month-Segment FE	Y	Y	Y	Y

Notes: This table displays regressions of share-weighted log prices for each molecule on the policy roll-out variable constructed using the first decree deadline. The average is taken over all drugs within each market. Panel A shows results by whether the drug treats a chronic condition, following the classification in Huber *et al.* (2013). Panel B shows results by whether the drug has high mortality effects, following the classification in Chandra *et al.* (2024). The sample size in columns 2–4 corresponds to the number of market-month observations with at least one product in the segment. We also drop markets without products in the segment at baseline (January 2010). Standard errors clustered at the market level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.